What an LLM Really Is

The autocomplete that read the internet

Module 1 · Zero to LLM

The ladder

- **AI** — any behaviour we'd call "smart"
- **Machine learning** — learn rules from examples
- **Deep learning** — many stacked layers learn their own features
- **LLM** — deep learning aimed at *one* job: **predict the next token**

> Spam filter: you don't write rules, you show 10 000 labelled emails.

Why "large"?

Model	Parameters (approx.)
Our bigram toy	a few hundred counts
GPT-2 small (we rebuild it)	~124 million
Frontier models	reported 100s of billions

Parameters = adjustable numbers. Training = nudging them.

The one loop

```
text → tokens → numbers → model → scores → probabilities → pick → append ↺
```

1. Chop text into **tokens**
2. Look each token up as a **vector** (M04)
3. **Mix** information between tokens — attention (M05)
4. Score **every** vocabulary entry (~50k for GPT-2)
5. **Pick** one (greedy vs sampling) and repeat

A chat is just the loop

- Your message + history is pasted in front
- No memory between chats unless the app re-sends it
- No fact database — learned patterns

Live: the dumbest LM

Bigram counting in 20 lines → `snippets/01_bigram_counts.py`

Four honest limits

Limit	Falls out of…
Hallucination	the loop *always* emits a fluent token
Letter counting / maths	it sees tokens, not letters
Knowledge cutoff	frozen training data
Context limit	fixed window

Recap

Next-token predictor **+ scale**.

Next: **M02 — the 20 % of Python & PyTorch you need**